

Impact of transaction pricing mechanisms on energy community benefits sharing

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Abstract—When the price of the energy shared within an energy community is based on a local energy market, it is the responsibility of each participant to bid adequately so that participating provides a larger benefit than not participating. Alternatively, centralized energy community bill minimization may be an option, but a mechanism to share the collective benefits among the members is needed. This mechanism should be fair and easy to explain, no members should be harmed with respect to their individual optimal behavior and should provide the right economic signal. This paper analyses and compares some common pricing mechanisms for the internal compensation for the energy shared among the members of an energy community centrally managed. Simple case examples are used to identify those pricing mechanisms that are fairer and provide the righter economic signals to the participants.

Index Terms—renewable energy communities, energy sharing, pricing mechanisms, collective benefits sharing.

I. INTRODUCTION

Renewable energy communities (REC) are based on the collective self-consumption (CSC) concept, which means that REC members can produce their own energy to satisfy their supply needs, but also share their energy surplus with other REC members needing supply. When the self-consumption is done with generation behind the meter, the meter readings already discount this generation from the consumption, and the retailer only bills for the real supply. However, when the energy self-consumed comes from external generation facilities or from other REC members, regulatory frameworks usually define the so-called allocation coefficients (AC) that allow to compute the share of energy allocated to each REC member. The REC manager computes and sends the AC to the distribution system operator (DSO) or the metering responsible entity, that uses the AC to compute and inform the retailers on the amount of energy self-consumed by each REC member, so the retailer only bills for the energy supplied by or injected to the main grid. Since the energy allocated to REC members can belong to others' members external facilities or surplus, these owners should be financially compensated for this energy.

If the allocated energy comes from the cleared REC transactions in an internal electricity market (LEM), then the resulting amounts and prices are the result of the participants' bids. However, local markets involve complexities that many

members may not be willing to adopt, disincentivizing the participation in CSC structures. Several LEM designs were reviewed in [1] including works on LEM that uses CSC regulation. Those that don't use market mechanisms to set prices were classified as system-determined mechanisms. Amongst them are fixed prices, prices determined by some sort of demand functions, or centralized cooperative determined games. In [2] some of the main pricing mechanisms proposed in the literature are described and analyzed qualitatively, and a new pricing mechanism based on the post-delivery pool concept is also assessed.

Reference [3] proposes a centralized REC optimization in two stages. In the first stage, the individual minimum energy cost of each REC member is computed. In the second stage, the collective cost is minimized with a centralized optimization approach [3, 4], but constrained to guarantee that no member has an energy cost larger than its individual optimum. To do so, a transaction price must be defined to compute the individual cost of the second stage to compensate for internal energy allocations. If this price is not fair enough, the constrained optimization limits the internal transactions so that no member is disadvantaged relative to its individual optimum.

This work simulates the pricing mechanisms reviewed in [2] and compares them considering a REC with storage systems and CSC allocation rules, assessing their behavior in the simulated scenarios. The paper is organized as follows. Section II describes the REC operation procedure and pricing mechanism considered. Section III describes case study and scenarios analyzed and section IV concludes.

II. ENERGY SHARING AND PRICING MECHANISMS

A. Energy sharing methodology

We assume that each REC member can have consumption, generation, and batteries all behind the meter (BTM). Each member has its own retail buying and selling tariffs, and the internal transactions are all compensated, for each interval settlement period (ISP) t (e.g., 60 minutes), at the same price λ_t^{p2p} for all REC members. As can be seen in Figure 1, REC member n can buy the electricity $E_{n,t}^{SUP}$ supplied by its retailer at price $\lambda_{n,t}^{buy}$, sell its surplus $E_{n,t}^{SUR}$ to its retailer at $\lambda_{n,t}^{sell}$, or buy $E_{n,m,t}^{PUR}$ internally from a REC member m (who sells $E_{m,n,t}^{SALE}$ to n)

at the internal price λ_t^{p2p} . Without loss of generality, we assume a Portuguese-like self-consumption regulation is in place.

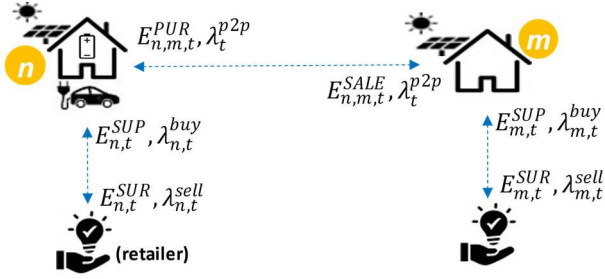


Figure 1: Electricity exchanges in a REC

A two-stages optimization procedure based on [1, 3] is used for the REC management. In this procedure, the second stage is the following centralized collective minimization of the total REC energy cost:

$$\min \sum_{t \in T} \left(\sum_{n \in N} (E_{n,t}^{SUP} \cdot \lambda_{n,t}^{buy} - E_{n,t}^{SUR} \cdot \lambda_{n,t}^{sell} + E_{n,t}^{SC} \cdot \lambda_t^{grid}) \right) \quad (1)$$

subject to the standard constraints described in [3] related to energy balances and the batteries operation, and where the last term of (1) refers to the grid tariffs for the self-consumed energy, given by:

$$E_{n,t}^{SC} = \min \left(\max(E_{n,t}^{CMET}, 0), \max \left(\sum_m (E_{n,m,t}^{PUR} - E_{n,m,t}^{SALE}), 0 \right) \right) \quad (2)$$

where $E_{n,t}^{CMET}$ is the metered energy of REC member n , positive for net consumption and negative for net injection. Stage 1 is computed as in (1) but for each individual REC member, and used in stage 2 to guarantee that the individual cost that results from the collective optimization is always equal or lesser than the cost C_n^{ind} that results from the individual behavior (where no energy sharing takes place):

$$\sum_t \left(E_{n,t}^{SUP} \cdot \lambda_{n,t}^{buy} - E_{n,t}^{SUR} \cdot \lambda_{n,t}^{sell} + E_{n,t}^{SC} \cdot \lambda_t^{grid} + \sum_m (E_{n,m,t}^{PUR} - E_{n,m,t}^{SALE}) \cdot \lambda_t^{p2p} \right) \leq C_n^{ind} \quad (3)$$

In practice, this constraint limits the energy shared when unfair prices lead to unfair financial compensation among REC members, as shown in [3].

To establish a pool-like market the following constraint is considered:

$$\sum_{n,m} (E_{n,m,t}^{PUR} - E_{n,m,t}^{SALE}) = 0 \quad (4)$$

Finally, the AC sent by the REC to the DSO specifying how the energy locally injected should be allocated among each REC member n are given by:

$$\alpha_{n,t} = \frac{\max(-E_{n,t}^{CMET}, 0) + \sum_m (E_{n,m,t}^{PUR} - E_{n,m,t}^{SALE})}{\sum_n (\max(-E_{n,t}^{CMET}, 0))} \quad (5)$$

AC allow the DSO to inform the retailer on the part of the consumption metered which is supplied from the main grid at the retailer tariff, or on the part of the injection sold back to the

grid. Note that, if no negative AC are allowed, then the following constraint must be added:

$$\sum_m (E_{n,m,t}^{SALE} - E_{n,m,t}^{PUR}) \leq \max(-E_{n,t}^{CMET}, 0) \quad (6)$$

B. Pricing mechanisms

From [2] the following pricing mechanisms are analyzed to compute the internal transactions price λ_t^{p2p} :

- Mid-market-rate (MMR):

$$p_h = \frac{\lambda_t^{buy} + \lambda_t^{sell}}{2} \quad (7)$$

where:

$$\lambda_t^{sell} = \max_n (\lambda_{n,t}^{sell}), \quad \lambda_t^{buy} = \min_n (\lambda_{n,t}^{buy}) \quad (8)$$

- Supply and demand ratio (SDR):

$$\text{if } 0 < sdr_h < 1 \text{ then } p_h = \frac{\lambda_t^{buy} \cdot (\lambda_t^{sell} + \lambda)}{(\lambda_t^{buy} - \lambda_t^{sell}) \cdot sdr_h + \lambda_t^{sell}} \quad (9)$$

if $sdr_h \geq 1$ then $p_h = \lambda_t^{sell}$

where:

$$sdr_h = \frac{eq_h}{dq_h} \quad (10)$$

being eq_h the total net local energy injection of all REC members and facilities, and dq_h the total net local energy consumption, both measured at the REC members' and facilities' meters.

- Compensated supply and demand ratio (SDRC):

if $0 < sdr_h < 1$ then

$$p_h = \frac{\lambda_t^{buy} \cdot (\lambda_t^{sell} + \lambda)}{(\lambda_t^{buy} - \lambda_t^{sell} - \lambda) \cdot sdr_h + \lambda_t^{sell} + \lambda} \quad (11)$$

if $sdr_h \geq 1$ then $p_h = \lambda_t^{sell} + \frac{\lambda}{sdr_h}$

where λ is a compensation factor such that $0 < \lambda < (\lambda_t^{buy} - \lambda_t^{sell})$ to still incentivize internal trades when the REC has a net surplus [2].

- Post-delivery pool (POOL):

As proposed in [2], the aggregated selling and buying curves of the post-delivery pool can be built from the actual energy delivered by the REC members and their opportunity costs with their retailers. For each ISP, when injecting, the REC member will have a selling bid with the energy injected and its retailer buying price, and when consuming, a buying bid with the energy consumed and its retailer selling price. Note that a prosumer will only have either a buying or a selling bid per ISP. By intersecting the aggregated selling and buying bids the pool equilibrium is computed which provides the market price. TABLE I summarizes the general benefits, challenges, and complexities of each mechanism.

TABLE I. COMPARISON OF PRICING MECHANISMS

MMR	✓ Readily computable and easily explainable. ✓ Intends to equally share the economic benefit. ↓ Not applicable if $\lambda_{n,t}^{sell} > \lambda_{n,t}^{buy}$ ↓ The definition of the denominator = 2 is arbitrary and may not represent the fairest choice.
SDR	✓ Readily computable. ✓ Can be fairer than MMR since it emulates the fundamental law of economics that assumes an inverse relationship between price and sdr_h. ✓ By applying a compensation when there is a generation surplus ($sdr_h > 1$), internal transactions can be incentivized over selling to the grid. ↓ The compensation factor is not directly computed from pre-known data. ↓ Can lead to higher REC overall costs than MMR.
POOL	✓ Easily explainable. ✓ Potential to be the fairest and to provide the lowest REC overall costs. ↓ Post-delivery mechanism (although, predictive optimization can still be applied if the algorithm considers shadow values).

III. SCENARIOS, SIMULATIONS AND RESULTS

A. Case study and simulation scenarios

A REC with four members P1, P2, P3 and P4 was used for the assessment. As shown in Figure 2, all members have BTM consumption, with some also having BTM storage and others with BTM renewable generation (PV panels). All transactions among the members use the public grid with a self-consumption grid access tariff λ_t^{grid} set to 0.016€/kWh.

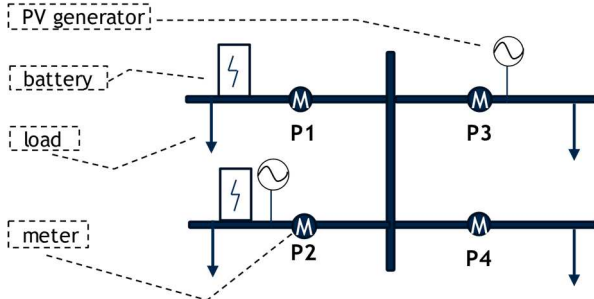


Figure 2: REC case example

Solar PV generation and load demand profiles were taken from [5], and scaled to 15 kWp, as shown in Figure 3. The batteries' capacity was set to 10 kWh with charging and discharging power of 2.5 kW and 90% of efficiency, and minimum and maximum state of charge (SoC) of 10% and 90% respectively. It was also assumed that their initial and final SoC for every scenario was 50%. Four different opportunity costs scenarios of the peers with their retailer or balance responsible party were defined according to TABLE II: ScenEQ represents a scenario where all members have equal buying and selling prices; Scen1 a scenario where producers have worse prices than consumers; Scen2 is the opposite of Scen1; Scen3 is a scenario where members with storage have worse prices than other members; and Scen4 is the opposite of Scen3.

We considered five pricing mechanisms for all scenarios, namely POOL, MMR, SDR and SDRC with two compensation strategies, SDRC, with $\lambda = (\lambda_t^{buy} - \lambda_t^{sell})$ and SDRC_HALF with $\lambda = 0.5 \cdot (\lambda_t^{buy} - \lambda_t^{sell})$. An additional operation and

pricing mechanism was considered, POOL_DUAL, ignoring constraint (3) in the collective optimization and setting the price λ_t^{p2p} to the shadow values of constraint (4).

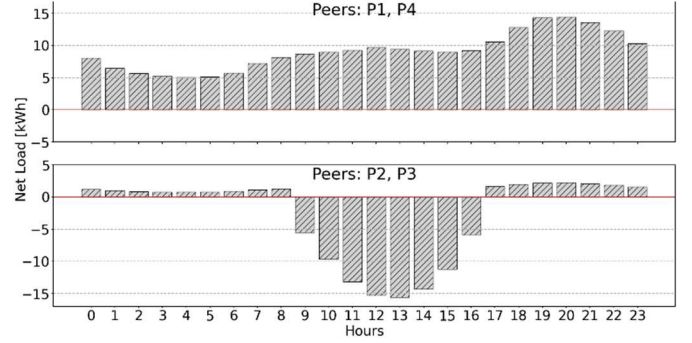


Figure 3: REC case example

TABLE II. PROSUMERS' BUYING AND SELLING TARIFFS (€/kWh)

Prosumer	ScenEQ	Scen1	Scen2	Scen3	Scen4
P1 ($\lambda_{n,t}^{sell}, \lambda_{n,t}^{buy}$)	0.04, 0.2	0.04, 0.2	0.04, 0.2	0.02, 0.4	0.08, 0.1
P2 ($\lambda_{n,t}^{sell}, \lambda_{n,t}^{buy}$)	0.04, 0.2	0.02, 0.4	0.08, 0.1	0.02, 0.4	0.08, 0.1
P3 ($\lambda_{n,t}^{sell}, \lambda_{n,t}^{buy}$)	0.04, 0.2	0.02, 0.4	0.08, 0.1	0.04, 0.2	0.04, 0.2
P4 ($\lambda_{n,t}^{buy}$)	0.2	0.2	0.2	0.2	0.2

For each scenario and pricing mechanism (except POOL_DUAL) an iterative process was performed to compute the final λ_t^{p2p} . At each step, λ_t^{p2p} is computed considering the current initial net load of each peer, as described in section II.B. Then, the two-step optimization procedure of [3], given λ_t^{p2p} , is performed, providing new values for the peers' net load due to the new battery schedules of peers P1 and P2. This loop is repeated, as described in [2], until the stopping criteria is reached, consisting in an price error tolerance or a maximum of 20 iterations. Indeed, when the Euclidean distance between the array λ_t^{p2p} at the current iteration and at the previous one is lower than 0.01€/kWh the algorithm finishes. In case the maximum number of iterations is reached, the solution corresponds to the iteration with the minimum REC cost.

The Mixed-Integer Linear Programming (MILP) was formulated in Python using puLP [6], and simulations were performed on an AMD Ryzen 5 PRO 4650U, 2.10 GHz, 16GB RAM laptop using the commercial solver GUROBI™ [7]. Since GUROBI™ does not provide the dual value of MILP constraints, the POOL_DUAL approach was computed using the COIN-OR CBC open-source solver [8] to compute those values and then use them as the array of λ_t^{p2p} prices when solving another MILP using GUROBI™.

B. Results and discussion

Figure 4 shows the total REC energy costs by scenario and by pricing mechanism. As expected and in line with [3], REC costs are always higher when no internal transactions are considered (Stage 1 scenarios) since there are no savings from collective self-consumption. It can also be seen that SDRC and SDRC_HALF consistently lead to higher energy costs, suggesting lower percentages of REC self-sufficiency, i.e., a higher dependence of members on their retailers. The MMR, POOL, and POOL_DUAL approaches provide the same prices

for all scenarios except for Scen4, where the cost of the latter has more than halved.

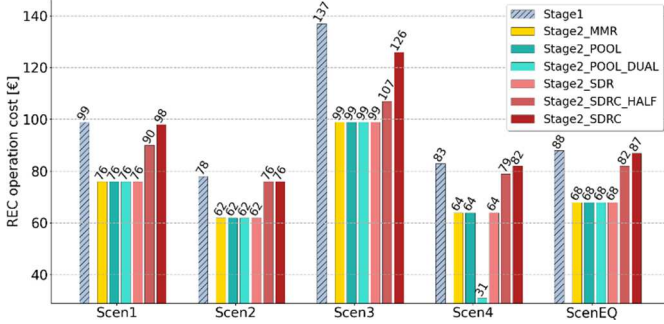


Figure 4: Total REC energy costs

Figure 5 shows the energy costs per peer considering all scenarios and pricing mechanisms. As can be seen, in Scen4 with POOL_DUAL simulation, peers without generation experience the largest cost reduction, while the other peers achieve comparable revenues to other mechanisms under the same scenario. The SDR approaches tend to be more inequitable, favoring producers (P2 and P3) who increase profits by increasing the consumption costs of purchasing peers (P1 and P4). This is particularly evident in Scen3 under SDR_HALF. It is also interesting to note the similarity of results for peers with the same net load profiles that differ in storage ownership. Comparing P1 with P4 and excluding Scen3 where $\lambda_{n=P1,t}^{buy} = 2 \times \lambda_{n=P2,t}^{buy}$, both peers achieve the same results, with small differences favoring P1 which owns the storage asset. On the other hand, P2 almost always achieves better costs/revenues than P3 in scenarios where the same prices and tariffs are applied to both. Of course, the potential of the storage asset is increased if a peer has its own generation, as it can charge the battery with its own surplus at zero cost.

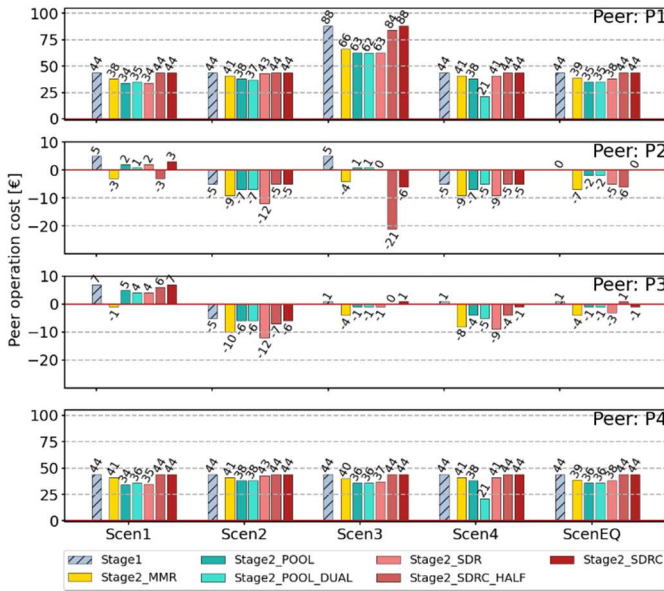


Figure 5: Individual REC members' energy costs

Figure 6 shows the prices that result from each simulated mechanism and for each scenario. For the hours with no consumption, 0 – 8 and 17 – 23, the prices vary significantly

depending on the chosen storage charge and discharge profiles. In fact, it was noticed that some scenario-mechanism combinations had difficulties in converging. Indeed, same values of total REC energy cost were achieved for different λ_t^{p2p} , so the price could change among iterations without impacting the cost, making the problem not determined. Within the hours of surplus generation, the prices stabilize and the differences among the different mechanisms are evident. Those with lower REC cost are those with lower prices, closer to the cost of the opportunity prices obtained in the POOL_DUAL simulation. The only exceptions occur in hours 9 and 16, where local production is lower than demand before storage dispatch is considered but becomes higher due to the final storage contribution. At these hours POOL_DUAL values are higher than SDR and MMR prices for all scenarios. This difference would be smaller if both mechanisms had been calculated considering only the offers accepted in the LEM pool and not all offers as was the case. Indeed, by doing so, the lowest buying bids would not have been considered for the price computation, increasing and bringing the calculated prices closer to the other mechanisms. Focusing once again on Scen4 although other mechanisms' prices closely follow the ones obtained in POOL_DUAL in hours with surplus, the price is halved for the other hours. Particularly at hours 1, 3 and 18 to 22, no other mechanism was able to capture this cost of opportunity.

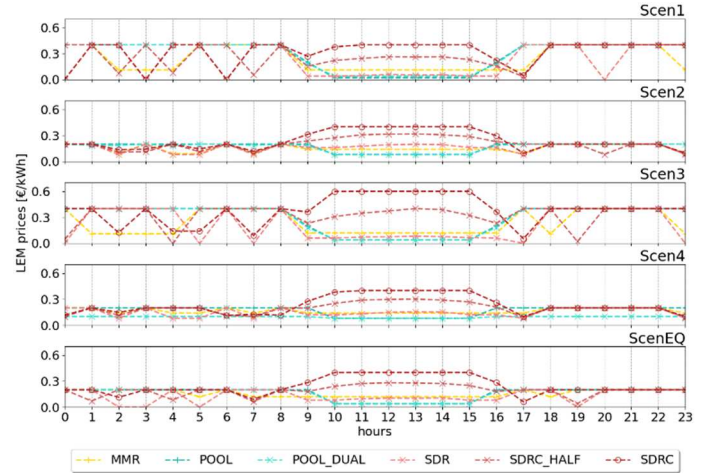


Figure 6: Internal transaction prices per scenario and mechanism

A more efficient REC operation comes from a less reliance on retailers and a higher proportion of consumption from local generation, as shows TABLE III. TABLE III shows some interesting and distinctive dynamics occurring between compensated SDR and the other types of mechanisms. Under SDR and SDR_HALF, peers with surplus (P2 and P3) preferably trade among themselves, a tendency that increases with the rise of the compensation. Other mechanisms encourage these transactions to be made with peers who have no generation. It can be concluded that, without a proper pricing mechanism, a REC can be uneven and fail to achieve its original purpose of increasing the self-sufficiency of communities, as already remarked in [3]. Note also that in Scen4, the reduced costs of peers P1 and P4 are due to the fact that all their needs are met locally.

TABLE IV shows the percentage of the surplus sold locally. There is a clear complementarity with TABLE III. Note that, despite having a storage unit, P1 never sells the stored energy on the local market, so it is not included in the table. In fact, from results not explicitly shown here, the surplus that P1 sells to the retailer is obtained exclusively in Scen1 by purchasing as much as possible (i.e., its net load under the positive coefficients constraint, see [3]) from peers P2 and P3, which have a very low feed-in tariff in this scenario, and selling it to the retailer.

TABLE III. PERCENTAGE OF PEER'S CONSUMPTION BOUGHT LOCALLY

		MMR	POOL	POOL_DUAL	SDR	SDRC	SDRC_HALF
Scen1	P1	38%	36%	36%	35%	4%	14%
	P2	0%	0%	0%	0%	61%	37%
	P3	0%	0%	0%	0%	62%	59%
	P4	35%	37%	37%	38%	5%	20%
Scen2	P1	39%	34%	37%	39%	5%	5%
	P2	0%	0%	0%	0%	60%	60%
	P3	0%	0%	0%	0%	63%	58%
	P4	32%	37%	33%	32%	2%	5%
Scen3	P1	37%	37%	37%	37%	18%	37%
	P2	0%	0%	0%	0%	51%	0%
	P3	0%	0%	0%	0%	59%	19%
	P4	28%	28%	28%	28%	0%	4%
Scen4	P1	37%	37%	100%	35%	6%	12%
	P2	0%	0%	0%	0%	61%	56%
	P3	16%	12%	100%	16%	62%	58%
	P4	32%	33%	100%	34%	1%	7%
ScenEQ	P1	35%	35%	34%	35%	6%	10%
	P2	0%	0%	0%	0%	62%	54%
	P3	0%	0%	0%	0%	64%	64%
	P4	31%	31%	31%	31%	3%	11%

TABLE IV. PERCENTAGE OF PEER'S SURPLUS SOLD LOCALLY

		MMR	POOL	POOL_DUAL	SDR	SDRC	SDRC_HALF
Scen1	P2	100%	100%	100%	100%	40%	75%
	P3	100%	100%	100%	100%	32%	36%
Scen2	P2	69%	69%	75%	90%	37%	35%
	P3	100%	100%	94%	78%	32%	38%
Scen3	P2	100%	100%	100%	100%	53%	100%
	P3	68%	68%	68%	68%	30%	13%
Scen4	P2	69%	69%	93%	69%	36%	41%
	P3	100%	100%	100%	100%	36%	48%
ScenEQ	P2	100%	73%	71%	100%	42%	58%
	P3	68%	92%	94%	68%	36%	36%

IV. CONCLUSIONS

This work compares various internal transaction pricing mechanisms to assess the incentives they provide to REC members, regardless of their role in the community (producers, consumers and/or storage owners). Indeed, fair local prices are essential for a fair sharing of the collective benefits the REC provides to its members, encouraging internal transactions and REC self-sufficiency by reducing the dependence of individual peers on their retailers in an equitable manner. Unfair prices trigger the constraint of not harming the individual benefits and significantly limit the local energy sharing.

Some of the conclusions are:

- There seems to be clear tendency for the compensated form of the SDR mechanism to benefit producers and penalise consumers.

- However, the pool, MMR and uncompensated SDR mechanisms provided similar results under the scenarios tested.
- Both mechanisms may benefit from a minor adaptation, considering that only the offers accepted in the LEM pool were used for their calculation. This implies considering, in each iteration, the impact of flexible resources on the total internal demand and supply of the LEM.
- The iterative process required by all mechanisms except POOL_DUAL, as well as the two-stage approach used to avoid undermining individual benefits, can be avoided by considering collective optimization with shadow prices as local prices. This approach achieved results as good as any other approach in most scenarios, and in a particular one it far outperformed the others, showing that in some cases there might be opportunity costs that are entirely missed by the other mechanisms, even though shadow prices can be controversial in problems with binary variables.

Current work is focusing on scaling this assessment to larger communities with more diverse price profiles and consider complementary business use cases, such as including shared resources and providing flexibility-related services to system operators.

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